1) z - Transform & ROC

a) $X(z) = 1 + 2z^{-1} + z^{-2} + 2z^{-3} + 3z^{-4}$

ROC : $|z| > 0$

b) $X(z) = \frac{z}{(z-2)^2}$, ROC : $|z| > 2$

c) $Log(n)$: No z-transform

d) $X(z) = \frac{z}{(z-2)} - \frac{z}{(z-0.5)}$

ROC = ϕ

e) $X(z) = \frac{z^{-6}}{(1-z)}$, ROC : $|z| < 1$

f) $X(z) = z^5$: ROC : whole z-plane

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Inverse z-transform

$h(n) = \delta(n) + \delta(n-1)$

$h(n) = z^{-1} - \frac{z^{-2}}{2} + \frac{z^{-3}}{3} + \frac{z^{-4}}{4} \ldots$

$h(n) = -2 \cdot 2^{n} u(n) + 3 \cdot 3^{n} u(n)$

$h(n) = 2(4)^{n} - 2^{n}$

$h(n) = 1/3 - \delta(n)$

LCCD = $y(n) = y(n-1) + 5x(n) + 2x(n-1)$

Impulse Response = $h(0) = 5$
$h(1) = 7$
$h(2) = 7 \ldots$

$$H(e^{j\omega}) = \frac{(5 + 2e^{-j\omega})}{(1 - e^{-j\omega})}$$

o/p for $x(n) = 2^{n} \cdot u(n)$
via $Y(z) = H(z) X(z)$

$$H(z) = \frac{z}{[(z-2)(z-3)]}$$

$$y(n) = 5y(n-1) - 6y(n-2) + x(n-1)$$

Unit step = use input $u(n)$
Resp.

$x(n) = 2^n u(n)$, $h(n) = 5^n u(-n-1)$

$X(z) = \frac{z}{(z-2)}$, $H(z) = \frac{z}{(z-5)}$

$Y(z) = X(z) \cdot H(z)$

DTFT for

$(n+1) a^n u(n) = 1/[1 - a e^{-j\omega}]^2$

$\sin(\omega n)/\pi n$ = rectangular in freq.

$\cos(\omega n)$ = impulses at $\pm \omega$

$e^{(j\omega n)} u(n) = \frac{1}{[1 - e^{(j\omega - a)}]}$

Rect window = sin function

$$x(n) = \left(\frac{1}{2\pi}\right) \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

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8) Linear : full length $(N+M-1)$
Circular: Modulo N. wrap
Padding to match length

9) $|H(e^{j\omega})| = 1$
phase from : $\arg(H(e^{j\omega}))$

10) Linearity : $ax_1 + bx_2 \rightarrow ax_1(z) + bx_2(z)$
Time shift : $x(n-k) \leftrightarrow z^{-k}X(z)$
Scaling : $a^n x(n) \leftrightarrow X(z/a)$

11) $x(n)$ Real = $X(e^{j\omega}) = X(e^{-j\omega})$
Magnitude is even
phase is odd.

12) $x(n) = (1,1,2,2)$, $h(n) = (1,2,3,4)$
L.C. length = 7
Circular mod(4)

13) $X(k)$: nonzero at $k = 1, 7$
Magnitude = 4, phase = 0

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14)	Stable	Causal	Linear	T.	Memory
1	✓	✓	✓	✓	✓
2	✓	✗	✓	✗	✓
3	✗	✓	✗	✓	✓
4	✗	✓	✗	✓	✓
5	✗	✓	✓	✓	✗

15) $x(n) = u(n) + 3\delta(n-1) + 4u(-n-3)$

$x(n-1)$: shift right

$x(2n)$: compress

$x(n/2)$: expand

$x^2(n)$: square

$x(-n)$: reflect

16)

Signal	Periodic	Period
$e^{[j\pi n/6]}$	✓	12
$\sin 3n$	✗	-
$e^{[jn/1.41]}$	✗	-
$\sin(\pi n)/\pi n$	✗	-
$\sin(3\pi n) + \cos(6\pi n)$	✓	2

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